Multiplying & Dividing Rational Expressions

Name ________________ · —

Formula bank **Factor toolbox:** GCF · difference of squares $a^2 - b^2 = (a-b)(a+b)$ · trinomials

Divide rule: $\frac{A}{B} \div \frac{C}{D} = \frac{A}{B} \cdot \frac{D}{C}$ (flip, then multiply)

Restrictions: every ORIGINAL denominator $\neq 0$ — and when dividing, the divisor's top too

VOCABULARY

Rational expression — a fraction whose top and bottom are ________________

Restriction — an x-value that makes an ORIGINAL ________________ equal zero

Reciprocal — the flipped fraction — dividing means multiplying by the ________________

EXAMPLE 1 · TYPE 1 — MULTIPLY

Simplify $\frac{x^2+5x+6}{x} \cdot \frac{x}{x+2}$ and state restrictions.

1. Factor every top and bottom: $x^2 + 5x + 6 =$ ________________
2. Cancel common factors: the $(x+2)$ pair and the _______ pair cancel
3. Simplified result: __________
4. Restrictions from ORIGINAL denominators: $x \neq$ __________

EXAMPLE 2 · TYPE 2 — DIVIDE (FLIP FIRST)

Simplify $\frac{2x^2+9x+4}{3x} \div \frac{2x+1}{x}$ and state restrictions.

1. Flip the divisor and multiply: $\div \frac{2x+1}{x}$ becomes · _____________
2. Factor: $2x^2 + 9x + 4 =$ __________________
3. Cancel $(2x+1)$ and x: result ___________
4. Restrictions (original denominators AND the divisor's top): $x \neq$ ____________

YOU TRY

1.

Simplify $\frac{8m}{m^2+5m} \cdot \frac{m}{4}$ and state restrictions.

2.

Simplify $\frac{3a}{a^2+a} \div \frac{3}{a}$ and state restrictions.

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